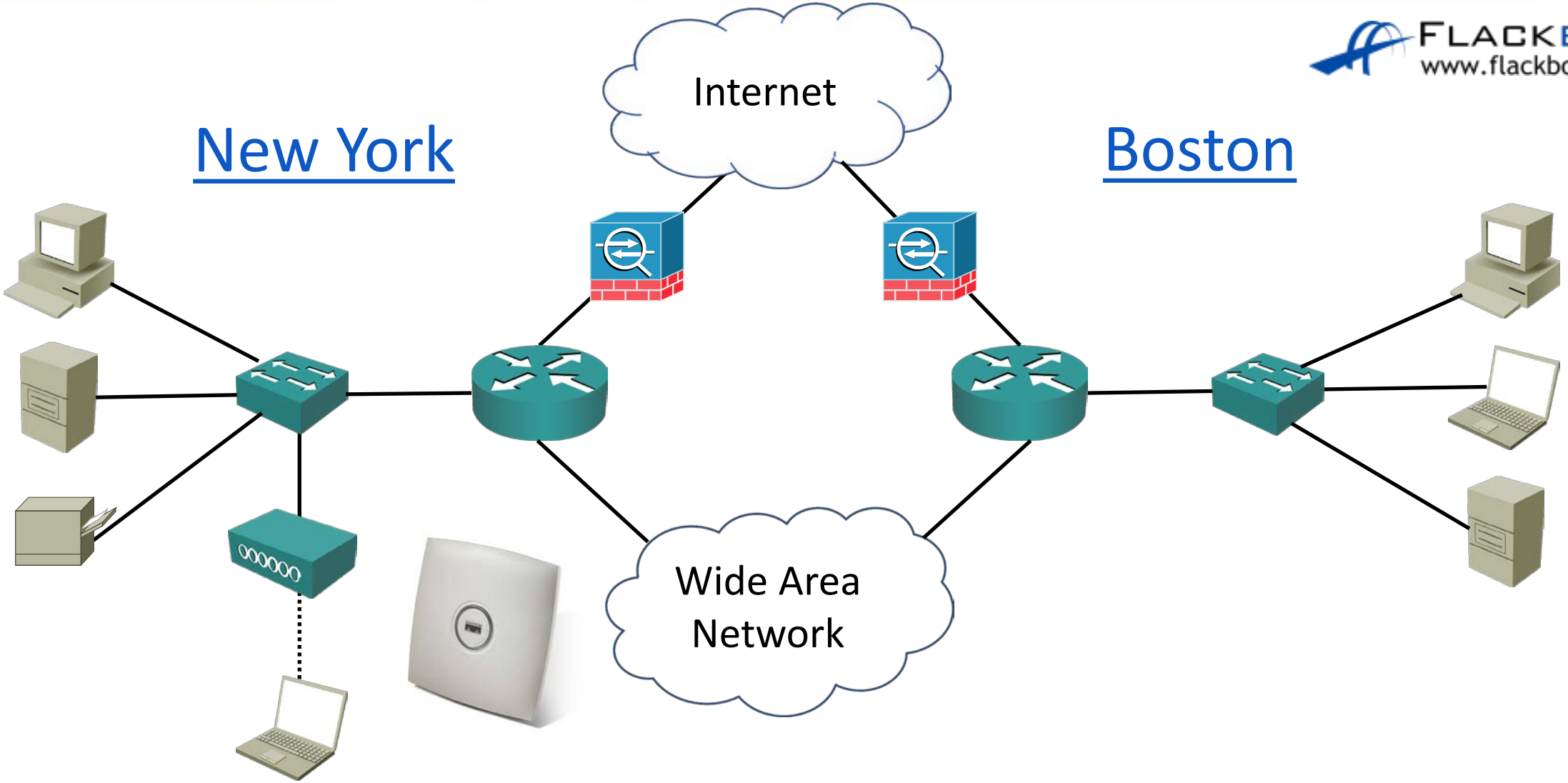
A (Very) Basic Introduction to Networking



FLACKBOX
www.flackbox.com

The OSI Open Systems Interconnect Model

- The OSI reference model is a standard of the International Organization for Standardization (ISO).
- It is a general-purpose framework that characterises and standardises how computers communicate with one another over a network.
- Its seven-layered approach to data transmission divides the operations into specific related groups of actions at each layer.
- A layer serves the layer above it and is served by the layer below it.

OSI Reference Model - Encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5			
4			
3			
2			
1			

L6 L7

OSI Reference Model - Encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4			
3			
2			
1			

L5	L6	L7

OSI Reference Model - Encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3			
2			
1			

OSI Reference Model - Encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2			
1			

L3	L4	L5	L6	L7

OSI Reference Model - Encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1			

L2	L3	L4	L5	L6	L7

OSI Reference Model - Encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		Hubs

L2	L3	L4	L5	L6	L7

OSI Reference Model – De-encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		

L2 L3 L4 L5 L6 L7

OSI Reference Model – De-encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		

L2 L3 L4 L5 L6 L7

OSI Reference Model – De-encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		

OSI Reference Model – De-encapsulation

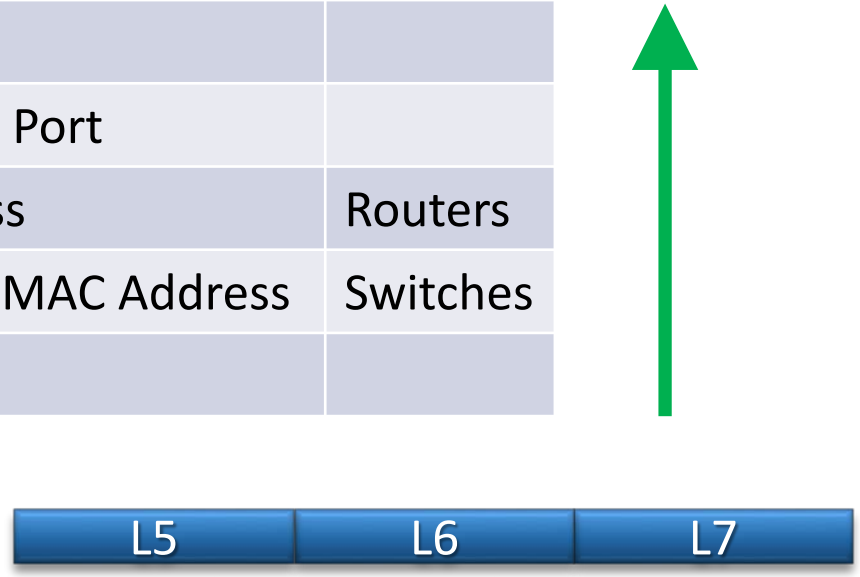
Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		

OSI Reference Model – De-encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		



OSI Reference Model – De-encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		

OSI Reference Model – De-encapsulation

Sender **Receiver**

Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		

L7

OSI Model Benefits

- Engineers do not need to design a technology to work end to end from top to bottom of the model. They can just focus on their layer of expertise, and make sure they comply with the standards for the layers above and below.
- This leads to open standards and multi-vendor interoperability.
- For example: If you’re an application developer, you can just focus on the top three layers, the lower layers are the domain of network engineers.
- Troubleshooting is easier because you can analyse a problem in a logical fashion layer by layer.

The OSI Open Systems Interconnect Model

- It's difficult to overstate how important the OSI Model is to computer networking.
- As you become more experienced you will 'think' according to the OSI model when you are troubleshooting or learning a new network technology.
- On the job you will hear technologies and problems being described according to their OSI layer.

OSI Acronyms

- The Classic: Please Do Not Throw Sausage Pizza Away
- Network Relevant: Please Don’t Need Those Stupid Packets Anyway
- Us Relevant: Please Do Not Teach Students Pointless Acronyms
- Useful: Please Do Not Take Sales People’s Advice
- My Favourite: Please Do Not Touch Superman’s Private Area

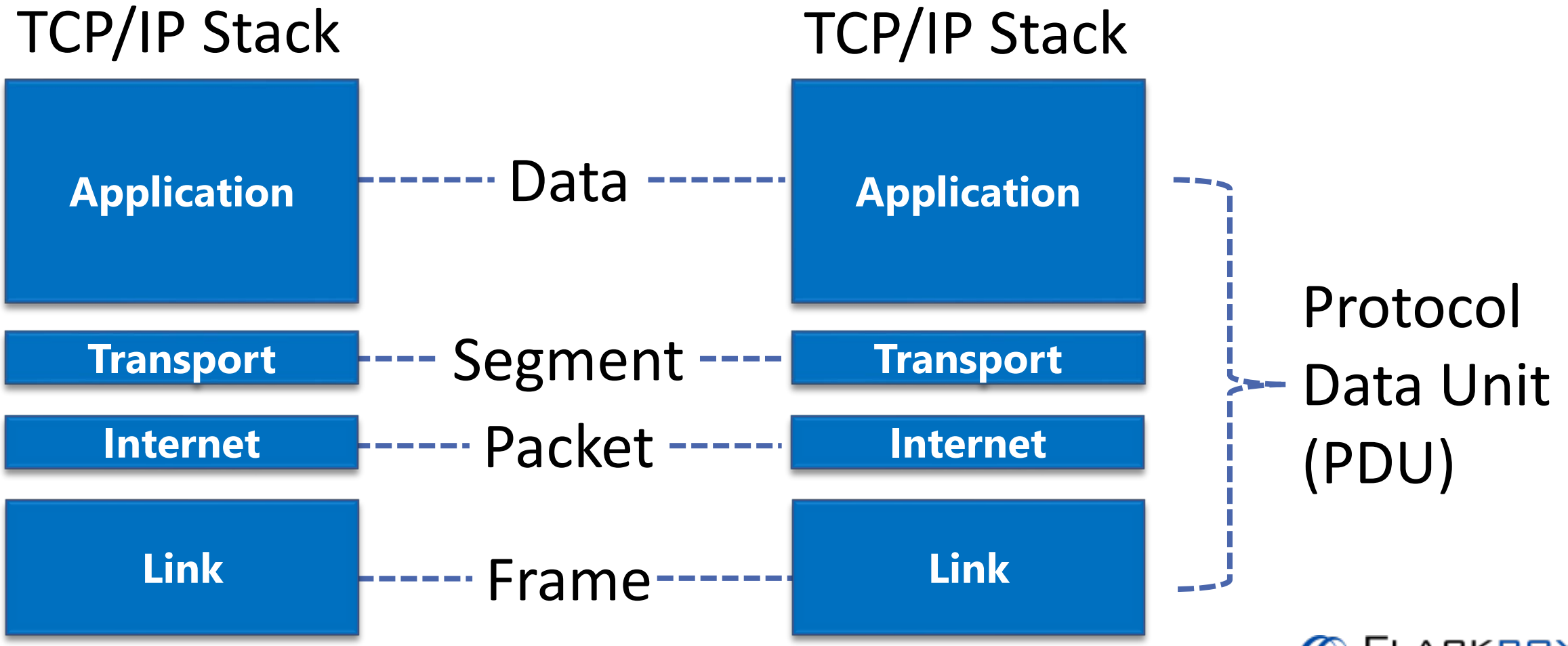
The TCP/IP Suite

- TCP/IP was developed during the 1960s by the US Department of Defense's (DoD) Advanced Research Projects Agency (ARPA).
- It is a protocol stack which consists of multiple protocols including TCP (Transmission Control Protocol) and IP (Internet Protocol).
- It is the main protocol stack used in computer operations today.
- Whereas the OSI Reference Model is conceptual, the TCP/IP stack is used to transfer data in production networks.
- TCP/IP is also layered but does not use all of the OSI layers, though the layers are equivalent in operation and function.

Comparing the OSI Model with the TCP/IP Stack

OSI Model	TCP/IP Stack	Cisco TCP/IP Stack Layer Definition
Application	Application	'Represents data users, encodes and controls the dialog.'
Presentation		
Session		
Transport	Transport	'Supports communication between end devices across a diverse network'.
Network	Internet	'Provides logical addressing and determines the best path through the network'.
Data Link	Link	'Controls the hardware devices and media that make up the network'.
Physical		

Host Communications Terminology



FLACKBOX
www.flackbox.com

The Upper OSI Layers

- Network engineers do not typically work directly with the upper 3 layers of the OSI model… but we still need to know what they do.
- They are more relevant to application developers.
- In this lecture I will primarily be giving you the Cisco definitions of the layers.
- Information included in the upper layers would include the Message Body and Subject Line in an email message for example.

Layer 7 – The Application Layer

- The application layer provides network services to the applications of the user.
- It differs from the other layers in that it does not provide services to any other OSI layer.
- The application layer establishes the availability of intended communication partners.
- It then synchronizes and establishes agreement on procedures for error recovery and control of data integrity.

Layer 6 – The Presentation Layer

- The presentation layer ensures that the information that is sent at the application layer of one system is readable by the application layer of another system.
- The presentation layer can translate among multiple data formats using a common format (eg computers with different encoding schemes).

Layer 5 – The Session Layer

- The session layer establishes, manages, and terminates sessions between two communicating hosts.
- The session layer also synchronizes dialog between the presentation layers of the two hosts and manages their data exchange.
- For example, web servers have many users, so there are many communication processes open at any given time to track.
- It also offers efficient data transfer, CoS, and exception reporting of upper layer problems.

The Lower OSI Layers

- Whereas Network engineers are not particularly interested in the upper OSI layers, we are **very** concerned with the lower 4 layers of the OSI model.
- Each of these layers have their own dedicated section later and you will learn much more detailed information about them throughout the course.

Layer 4 – The Transport Layer

- The main characteristics of the Transport layer are whether TCP or UDP transport is used, and the port number.
- Definition:
  - The transport layer defines services to segment, transfer, and reassemble the data for individual communications between the end devices.
  - It breaks down large files into smaller segments that are less likely to incur transmission problems.

Layer 3 – The Network Layer

- The most important information at the Network layer is the source and destination IP address.
- Routers operate at Layer 3.
- Definition:
  - The network layer provides connectivity and path selection between two host systems that may be located on geographically separated networks.
  - The network layer is the layer that manages the connectivity of hosts by providing logical addressing.

Layer 2 – The Data-Link Layer

- The most important information at the Data-Link layer is the source and destination layer 2 address.
- For example the source and destination MAC address if Ethernet is the layer 2 technology.
- Switches operate at Layer 2.
- Definition:
  - The data link layer defines how data is formatted for transmission and how access to physical media is controlled.
  - It also typically includes error detection and correction to ensure a reliable delivery of the data.

Layer 1 – The Physical Layer

- The Physical layer concerns literally the physical components of the network, for example the cables being used.
- Definition:
  - The physical link enables bit transmission between end devices.
  - It defines specifications needed for activating, maintaining, and deactivating the physical link between end devices.
  - For example, voltage levels, physical data rates, maximum transmission distances, physical connectors etc.